MD HUMAYUN

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ML Engineer | Software Development Engineer | C++ | DSA | System Design | GATE CS 93.8 Percentile

SUMMARY

M.Tech CS student at SVNIT Surat (GATE **93.8** percentile) working where machine learning meets systems engineering. Built an inline intrusion prevention system (IPS-ML) evaluated on **2.5M** network flows, where a false positive drops a real connection — so it is evaluated as a control system, not a classifier. Also built a multi-agent drone swarm simulator under a MeitY-sponsored project, with a paper under review at IEEE GSCON 2027, and production-style pipelines in C++ and Python across REST APIs, SQLite, Docker and AWS.

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT Surat) *2025 – Present*
M.Tech in Computer Science (Information Security & Privacy) | ***CPI: 8.70/10*** **Surat, Gujarat**

Bihar Engineering University *2020 – 2024*
B.Tech in Computer Science Engineering | ***CGPA: 8.51/10*** **Patna, Bihar**

PROFESSIONAL EXPERIENCE

Research Intern — Hybrid AI-Based Autonomous Drone Swarm Navigation May 2026 – Jul 2026
MeitY (Ministry of Electronics & IT), Govt. of India — Sponsored Project, SVNIT Surat (Guide: Prof. Mukesh A. Zaveri)
Python | Blender | LLMs (Groq) | LoRA | PPO | Gymnasium *Team project (2)* | [Demo Video]

- Designed **BattleWorld**, a multi-agent simulator wrapped as a Gymnasium environment with a uniform **12-dimensional** state interface, so every commander — classical or learned — consumed identical input and the benchmark measured policy rather than input.
- Built the scenario pipeline: procedural urban scenes in Blender auto-labelled through an LLM annotation stage, making scenario count a parameter instead of a bottleneck.
- Fine-tuned an LLM with LoRA on the generated tactical labels, reaching **91.2%** mean token accuracy at **0.463** training loss.
- Benchmarked **seven** commander architectures over **28** held-out scenarios; hybrid LLM+PPO reached **67.9%** action accuracy against **60.7%** for the best classical baseline.

Salesforce Developer Intern Dec 2023 – Jan 2024
DST (Digital Skills Training) — Remote

- Developed Apex triggers and automated workflows in OOP-based backend logic, cutting manual data-entry overhead by **35%**.
- Customised KPI dashboards for **5+** cross-functional teams, replacing manually compiled weekly reports.

PUBLICATIONS

Hybrid AI-Based Autonomous Drone Swarm Navigation | *Guided by Prof. Mukesh A. Zaveri, SVNIT Surat* *Under Review, IEEE GSCON 2027 (#71435)*

PROJECTS

IPS-ML / ShieldNet — Intrusion Prevention with Explainable AI (M.Tech Project) *Aug 2025 – Present*
Python | scikit-learn | XGBoost | SHAP | CICIDS2017 | pytest [GitHub]

- Trained six detectors and a soft-voting ensemble on **2,497,980** audited CICIDS2017 flows, evaluated on an identical held-out set of **749,394**; ensemble reached **0.8814** macro-F1 at a **0.00135** false alarm rate and **0.432 ms** median inference.
- Replaced argmax with a confidence-aware ALLOW / UNCERTAIN / BLOCK decision, choosing the operating point by sweeping the threshold against a stated **1%** false-alarm budget; routing **1.62%** of traffic to review cut missed attacks to **0.0149%** and benign disruption to **0.0486%**.
- Computed SHAP attributions inside the enforcement path and wrote them to an append-only audit log beside each action, so every block is explainable after the fact.
- Improved rare-class detection over the published Random Forest baseline without synthetic oversampling: SQL injection F1 **0.11** → **0.50**, Heartbleed **0.77** → **1.00**; covered by **29** unit tests with all thirteen plan deviations logged.

Multi-Source Job Intelligence Platform *Apr 2026 – Present*
Python | Playwright | SQLite | GitHub Actions | Streamlit | Groq [GitHub]

- Built a headless scraping pipeline across **11** job boards and **9** government portals, persisting **2,158** listings to SQLite with idempotent upserts and per-scraper retry isolation, so one broken source degrades that source instead of failing the run.
- Applied fuzzy deduplication via `SequenceMatcher` at a tuned **0.88** threshold with weighted skill scoring, reducing **793** raw records to **357** ranked matches — a **55%** noise cut.
- Made LLM enrichment optional rather than required: a Groq-hosted model re-scores listings when configured, with a rule-based scorer as fallback, so the pipeline never depends on a paid service to run.

- Orchestrated an **8-stage** pipeline on GitHub Actions cron, fanning out to Excel workbooks, HTML digests, email alerts and a Streamlit dashboard with no manual trigger.

DDoS Detection & Mitigation — SDN + Machine Learning *Aug 2025 – Jan 2026*

Python | Mininet | POX | OpenFlow | Random Forest | Scapy [GitHub]

- Implemented detection inside a POX SDN controller tracking per-host packet rate, installing OpenFlow `DROP` rules in real time so mitigation is a control-plane action rather than an external appliance.
- Trained a Random Forest on SDN traffic features to replace the fixed-threshold baseline, which misfires on legitimate traffic spikes.
- Measured impact with `iperf` and `ping` rather than asserting it: throughput fell **54.3 → 31 Gbps** and average latency rose over **10x** under an ICMP flood.

Medical Assistant — RAG Pipeline with Local Inference *Jan 2024 – May 2024*

Python | LangChain | Pinecone | Ollama | Flask | Docker | AWS EC2 [GitHub]

- Built ingestion and serving over a **1,152-page** corpus: **500**-character overlapping chunks, top-**3** retrieval, **768-dim** embeddings in a cosine Pinecone index, served by a LangChain chain behind Flask.
- Constrained the chain to retrieved context with an explicit refusal fallback at **temperature 0.4**, so an out-of-corpus question is refused instead of answered from parametric memory.
- Ran inference fully locally via Ollama (Llama 3.2, `nomic-embed-text`), removing third-party LLM API dependencies; containerised with Docker on AWS EC2.

Latent Space Analysis with Variational Autoencoders *Feb 2026 – Apr 2026*

Python | TensorFlow | Keras | NumPy | scikit-learn [GitHub]

- Reproduced Kingma & Welling's *Auto-Encoding Variational Bayes* in TensorFlow — encoder, decoder, reparameterisation trick and ELBO — then extended it to a Conditional VAE and convolutional **128×128** RGB faces.
- Diagnosed posterior collapse on the face model, traced it to a mis-scaled KL term, and verified recovery through latent mean and variance statistics rather than by inspecting samples, which look acceptable long before the latent space is healthy.
- Evaluated on **60,000** MNIST digits and **13,233** LFW faces, reaching **0.678** SSIM and **23.40 dB** PSNR on held-out data.

TECHNICAL SKILLS

Languages: C++ (primary), Python, MySQL, Bash
CS Core: Data Structures & Algorithms, System Design, OOP & Design Patterns, Operating Systems, DBMS, Computer Networks, Cryptography
ML / AI: TensorFlow, Keras, Scikit-learn, XGBoost, SHAP, LoRA, PPO, CNN, LSTM, VAE, Reinforcement Learning
LLM & Retrieval: LangChain, Pinecone, Ollama, Groq, RAG, Gymnasium
Web / Tools: Flask (REST API), Docker, Git & GitHub, GitHub Actions, AWS (EC2, S3), Linux, SDN, OpenFlow, Mininet

CERTIFICATIONS & ACHIEVEMENTS

GATE CS 2025: 93.8 Percentile (National Eligibility for M.Tech Admissions)
NPTEL: Cloud Computing & Ethical Hacking — **All India Rank 33**
Summer Research Internship Certificate: SVNIT Surat, MeitY-Sponsored Project, issued by Prof. M. A. Zaveri
IIT Bombay: Python for Data Science, Computer Fundamentals | **IIT Roorkee:** IoT & Embedded Systems
Community: Founded NITIAN RAHI — engineering education community of **52,000+** across YouTube, Facebook and Instagram